

INVENTION DISCLOSURE FORM

1. Title of Invention/Technology (brief & non-confidential):

Smart Glove for Sign Language

2. Abstract of the Technology (150 words only) :

A wearable Smart Glove system provides real-time sign language translation to help speech- and hearing-impaired people in India communicate better. The system includes five flex sensors, each conditioned with specific signal conditioning LM393 comparator modules with threshold tuning. These sensors connect to two synchronized ADS1115 analog-to-digital converter modules with different I2C addresses and forced initialization. This setup improves signal stability and reduces noise issues common with standard microcontroller ADCs. An MPU6050 inertial measurement unit with a Digital Low-Pass Filter and gyroscope offset calibration captures hand movements accurately. An ESP32 microcontroller handles sensor fusion using a complementary filter, normalizes finger bends in three states, and classifies gestures without needing to connect to the internet. Recognized gestures convert into text and speech in various Indian languages. The system includes a built-in Own Sign Language designed by the research team, with gesture sets documented in the accompanying research paper, enabling users to learn and communicate in selectable Indian languages without internet access.

3. Please briefly describe the problem that the technology solves and its advantages/benefits relative to competing technologies or products. What product might this invention become?

A key hardware challenge in existing embedded sign language gloves is the instability and poor resolution of the built-in ADC of microcontrollers such as the ESP32 when reading weak, non-linear analog signals from resistive flex sensors. This causes inconsistent gesture readings, reduced recognition accuracy, and unreliable real-time performance. Existing systems either ignore this issue or rely on computationally heavy filtering approaches that increase latency. The proposed system solves this through a dedicated per-sensor voltage amplification stage combined with dual external high-resolution ADS1115 ADC modules with synchronized acquisition, producing stable, repeatable flex sensor readings suitable for accurate gesture classification on an embedded platform.

Compared to camera-based sign language recognition systems, the proposed technology offers several advantages, including lower computational requirements, faster real-time response, improved portability, and better performance under varying environmental conditions such as lighting changes. The system supports both static gesture recognition and dynamic gesture-based word prediction, making communication more natural and efficient. In addition, the lightweight processing approach makes the system cost-effective and suitable for real-time embedded applications.

This invention can evolve into a commercial assistive communication product for daily use by deaf and mute individuals. In the future, it can be expanded into a multilingual sign-language translation system integrated with mobile applications, AI-based gesture prediction, cloud connectivity, and real-time voice assistance for healthcare, education, workplaces, and public communication environments.

4. Please briefly describe the objective of your inventive and the need it fulfills. The particular productivity it provides with its disclosure.

The main objective of the proposed invention is to develop a low-latency, offline, real-time wearable system for translating sign language gestures into text and speech on an embedded ESP32 platform. A key technical objective is to overcome the signal acquisition limitations and noise susceptibility of the ESP32's internal ADC by introducing a five-channel per-sensor operational amplifier-based signal conditioning stage in combination with dual externally addressed ADS1115 analog-to-digital converter modules configured with forced initialization and synchronized sampling.

A further objective is to achieve robust and stable dynamic gesture recognition through register-level configuration of the MPU6050 Digital Low-Pass Filter, gyroscope offset calibration, and complementary filter-based orientation estimation, selected to minimize computational overhead while maintaining high accuracy on resource-constrained hardware.

Kindly describe the importance and the advantages of your innovation

A technically significant advantage of this innovation is its dedicated per-sensor signal conditioning architecture. Five individual operational amplifier voltage amplification circuits one for each flex sensor address the low-voltage, non-linear output characteristics of resistive flex sensors, which prior systems typically leave unaddressed or compensate for through software alone. This hardware-level amplification, combined with two independently addressed ADS1115 ADC modules operating with forced initialization on a shared I2C bus, provides high-resolution, synchronized, and repeatable analog-to-digital conversion that is demonstrably more stable than direct ESP32 ADC sampling. This specific hardware pipeline **flex sensor** → **dedicated LM393 signal conditioning module** → **dual ADS1115** → **ESP32** has not been documented in existing sign language glove literature and represents a core technical contribution of this invention.

One of the major advantages of this innovation is its real-time performance and portability, making it suitable for daily use in educational institutions, workplaces, healthcare environments, and public spaces. Unlike traditional camera-based systems, the proposed approach is less affected by lighting conditions and environmental disturbances, resulting in more reliable gesture recognition. The system is also designed to be cost-effective, lightweight, and user-friendly, which increases accessibility for a wider population.

Another significant advantage is its ability to support both static gesture recognition and dynamic gesture-based communication, improving the natural flow of interaction. The innovation also provides scalability for future developments such as multilingual translation, mobile application integration, AI-assisted prediction, and cloud-based communication support, making it a strong foundation for next-generation assistive communication technologies.

5. Components used in the making of your system

- ESP32 Microcontroller - selected for its dual-core processing, I2C/SPI support, Wi-Fi/Bluetooth capability, and suitability for lightweight embedded ML inference
- Flex Sensors (5 units)
- LM393 Soil Moisture Sensor Comparator Modules (5 units) — complete signal conditioning boards, one dedicated per flex sensor, repurposed from soil moisture sensing application for flex sensor signal conditioning; each module includes LM393 IC, onboard potentiometer for threshold adjustment, power LED, and D0 indicator LED
- ADS1115 ADC Modules (2 units) — 16-bit external ADC modules configured with distinct I2C addresses (0x48 and 0x49) for simultaneous dual-module operation on a single I2C bus with forced initialization for synchronized sampling
- MPU6050 IMU Sensor - 6-axis accelerometer and gyroscope for capturing hand orientation (pitch, roll, yaw); configured via register-level DLPF programming for noise reduction
- Breadboard / PCB Connections
- Jumper Wires and Connecting Cables
- Power Supply / Battery Module
- Smart Glove Fabric Base

Here kindly describe the composition, the concentration and the require elements in making the inventive product.

The proposed inventive product is composed of five functional layers integrated into a wearable glove platform: Sensing Layer: Five resistive flex sensors are mounted along each finger of the glove. Each sensor produces a weak, non-linear analog voltage signal proportional to the degree of finger bending. One MPU6050 IMU sensor captures three-axis acceleration and three-axis gyroscopic data representing dynamic hand orientation and motion.

Signal Conditioning Layer: Five complete LM393 comparator modules one per flex sensor condition and stabilize the weak analog outputs before digitization. Each module's onboard potentiometer is individually tuned to the bending threshold of its corresponding flex sensor. This per-sensor dedicated conditioning architecture, repurposed from soil moisture sensor design, ensures signal levels are within the optimal input range of the external ADC modules, reducing noise and improving measurement repeatability.

Data Acquisition Layer: Two ADS1115 16-bit ADC modules, configured with distinct I2C addresses, perform synchronized high-resolution analog-to-digital conversion of all five amplified flex sensor channels. Forced initialization of both modules ensures proper synchronization on startup. The MPU6050 is read simultaneously over the same I2C bus.

Preprocessing and Classification Layer: The ESP32 applies gyroscope offset calibration, register-level DLPF configuration on the MPU6050, and complementary filter-based orientation estimation to produce stable pitch and roll values. Flex sensor readings are normalized into three discrete states per finger — straight, half-bent, and fully bent — using per-sensor threshold calibration. These features are fused into a gesture feature vector for lightweight classification.

Output Layer: Recognized gestures are mapped to text and speech outputs in selectable Indian languages. An Own Sign Language mode allows users to record, store, and map personalized gesture templates to custom text or audio outputs, operating fully offline without cloud dependency.

6. Working Principle of your product along with the Detailed technical description of the working of each component and the working of the entire system fulfilling each objectives.

Existing sign language glove systems suffer from unreliable analog signal acquisition, unsynchronized multi-sensor sampling, and high computational overhead in motion processing when implemented on resource-constrained microcontrollers. These limitations result in inaccurate gesture recognition and increased latency.

The proposed Smart Glove system is developed as a five-stage hardware-software pipeline specifically designed to overcome the signal instability, noise, and latency challenges of existing embedded sign language glove systems. The pipeline consists of: Sensor Layer → Signal Conditioning Layer → Data Acquisition Layer → Preprocessing Layer → Classification and Output Layer.

Sensor Layer

The system employs five resistive flex sensors — one mounted along each finger of the glove and one MPU6050 six-axis IMU sensor mounted on the back of the hand.

The flex sensors produce analog resistance variations proportional to the degree of finger bending. These analog outputs are inherently weak and non-linear, making direct microcontroller ADC reading unreliable a limitation not addressed in most prior glove systems.

The MPU6050 provides three-axis accelerometer data and three-axis gyroscope data, enabling detection of dynamic hand orientation including pitch, roll, and yaw. This allows the system to recognize not only static finger-position gestures but also dynamic motion-based word-level gestures.

Signal Conditioning Layer

The raw analog output of each flex sensor is weak, non-linear, and highly susceptible to noise, making it unsuitable for direct sampling using the ESP32's internal ADC, which suffers from non-linearity and instability at low voltage levels. To address this, the system integrates five dedicated LM393-based signal conditioning modules, one for each flex sensor.

Each module, originally designed for resistive soil moisture sensing, is repurposed to condition flex sensor signals due to the similarity in resistive sensing behavior. The onboard potentiometer of each module is individually calibrated to match the bending characteristics of its corresponding finger, enabling per-sensor threshold tuning and improved consistency.

These modules stabilize and condition the sensor outputs before digitization, improving signal-to-noise ratio, reducing non-linearity effects, and ensuring reliable input to the ADS1115 ADC modules. This per-sensor LM393-based conditioning architecture represents a novel and practical hardware solution not commonly implemented in existing sign language glove systems.

Data Acquisition Layer

The five amplified analog signals are digitized using **two ADS1115 16-bit analog-to-digital converter modules**. The ADS1115 offers significantly higher resolution and stability compared to the ESP32's built-in 12-bit ADC, which suffers from non-linearity errors especially at lower voltage ranges.

The two ADS1115 modules are configured with distinct I2C addresses — 0x48 and 0x49 — enabling simultaneous dual-module operation on a single shared I2C bus. **Forced initialization** is implemented at system startup to ensure both modules are properly synchronized before any gesture data is collected, preventing read failures or desynchronization during operation.

The ESP32 manages all I2C communication — reading digitized flex data from both ADS1115 modules and simultaneously reading motion data from the MPU6050 over the same bus.

Preprocessing Layer

1. Hardware-Level Preprocessing — IMU Startup Calibration (Novel Contribution)

A key novel feature of this system is a **5-second IMU startup calibration routine** executed each time the glove is powered on. During these 5 seconds, the user holds the glove still in their natural resting position. The system collects 2000 gyroscope samples and computes the mean offset for each axis:

$$\text{Offset: } \Sigma \frac{\text{gyro readings}}{N} \text{ N:Calibration Sample, N= 2000}$$

This calibration is position-independent — meaning the system calibrates from **whatever angle or orientation the glove is held at startup**, not from a fixed predefined position. All subsequent orientation calculations and gesture predictions are made relative to this user-specific, session-specific baseline. This makes the system robust to variations in how different users naturally hold their hands, eliminating the need for users to adopt a fixed calibration pose.

This approach is more practical for real-world use compared to systems requiring fixed-position calibration, and has not been documented in existing sign language glove prior art in this specific form.

2. Hardware-Level Preprocessing — IMU Filtering

Several filtering approaches were evaluated for MPU6050 output stabilization including low-pass, Kalman, Madgwick, and Mahony filters. All of these introduced measurable computational overhead and latency on the ESP32 platform. To preserve the balance between accuracy and processing efficiency, a complementary filter was selected:

$$\text{Angle } (\theta) = 0.98 \times (\text{prev_angle} + \text{gyro} \times \text{dt}) + 0.02 \times \text{accel_angle}$$

This filter combines the short-term accuracy of gyroscope data with the long-term stability of accelerometer data at minimal computational cost — critical for real-time embedded operation. Additionally, the MPU6050's internal Digital Low-Pass Filter (**DLPF**) was configured directly via register-level programming using the Wire library, and gyroscope scaling was adjusted to improve measurement precision.

3. Software-Level Preprocessing — Flex Sensor Normalization

At the software level, each flex sensor's output is normalized into three discrete bending states:

- Straight
- Half-bent
- Fully bent

Per-sensor minimum and maximum threshold values are calibrated individually to account for variations between sensors and between users. This three-state normalization ensures consistency in gesture interpretation across different users and usage sessions.

The final feature set combines the five discretized flex sensor states with the MPU6050 orientation values (roll and pitch), enabling the system to distinguish both static alphabet gestures and dynamic motion-based word-level gestures.

Feature Extraction and Classification Layer

The preprocessed data is fused into a structured feature vector consisting of:

- Five normalized flex sensor bend states (straight / half-bent / fully bent)
- MPU6050 roll angle
- MPU6050 pitch angle

This feature vector represents the complete gesture pattern and is passed to a lightweight classification model. The classification algorithm is selected specifically for low computational cost to ensure real-time inference on the ESP32 without external processing dependencies.

Output Layer

Upon successful gesture recognition, the system maps the identified gesture to predefined text and speech outputs. The recognized sign is displayed as readable text and simultaneously converted to audible speech through a text-to-speech component in a selectable Indian language.

The system supports a built-in Own Sign Language (OSL) gesture set— a custom gesture language designed by our team and documented in the accompanying research paper. OSL gesture templates are pre-stored in the ESP32's classification library and mapped to text and speech outputs in selectable Indian languages, operating fully offline without cloud connectivity.

Describe the making and the working of the product in detail.

The Smart Glove is constructed as a fully standalone wearable assistive communication device built across five integrated hardware stages. Each stage is designed with specific component choices justified by technical requirements of reliable real-time embedded gesture recognition.

Making of the Product

Glove Base and Sensor Mounting: A fabric glove serves as the wearable base. Five resistive flex sensors are mounted along each finger, secured to follow natural finger bending motion accurately during gesture performance.

Signal Conditioning Stage: The raw analog output of each flex sensor is weak, non-linear, and susceptible to noise making it unsuitable for direct microcontroller ADC sampling. The ESP32's built-in ADC is known to exhibit non-linearity errors, voltage reference instability, and a noisy floor particularly at lower voltage ranges, causing inconsistent and unrepeatable flex sensor readings when connected directly.

To solve this, five complete LM393 comparator modules are integrated into the system one dedicated module per flex sensor. Each module is a complete signal conditioning board originally designed for soil moisture sensing applications, comprising an LM393 dual comparator IC, an onboard potentiometer for reference voltage adjustment, a power indicator LED, a D0 output indicator LED, and VCC, GND, and output pins.

The key novelty here is the repurposing of soil moisture sensor LM393 comparator modules for flex sensor signal conditioning in a sign language glove system. Soil moisture sensors operate on the same principle as flex sensors both are resistive sensors producing weak, variable analog voltages proportional to a physical quantity. The LM393 module was designed to condition exactly this type of weak resistive sensor signal, making it directly applicable to flex sensor outputs.

Each of the five LM393 modules serves the following function in the pipeline:

- Receives the weak analog voltage output from its dedicated flex sensor
- The onboard potentiometer is individually tuned per module to set the reference voltage corresponding to the bending threshold of that specific finger's flex sensor
- Stabilizes and conditions the signal before it reaches the ADS1115 ADC input
- The D0 output indicator LED provides a visible real-time confirmation that the module is actively reading its sensor useful during calibration and testing

Each module's potentiometer was individually calibrated to account for physical variation between the five flex sensors, ensuring consistent signal conditioning across all five channels.

This five-module dedicated per-sensor LM393 signal conditioning architecture, repurposed from soil moisture sensor design and applied to flex sensor conditioning in a wearable sign language glove, has not been documented in existing sign language glove literature and constitutes a core hardware novelty of this invention. Prior art systems either connect flex sensors directly to microcontroller ADC pins suffering from ESP32 ADC instability or use generic op-amp amplifier circuits without the integrated threshold adjustment and visual feedback features present in this implementation.

Data Acquisition Stage: The five conditioned signals from the LM393 modules are fed into two ADS1115 16-bit ADC modules. The ADS1115 was chosen over the ESP32's built-in ADC specifically because the ESP32 ADC suffers from non-linearity and noise issues at lower voltage ranges. The ADS1115 provides 16-bit resolution, internal precision voltage reference, and a programmable gain amplifier all critical for reliably distinguishing the three bending states per finger.

The two modules are assigned distinct I2C addresses — 0x48 and 0x49 — allowing simultaneous dual-module operation on the single shared ESP32 I2C bus. Module 1 handles three flex sensor channels and Module 2 handles the remaining two. Forced initialization is implemented at every startup to explicitly verify both modules are responding correctly before any gesture data is collected, preventing silent read failures.

IMU Sensing and Mounting: An MPU6050 six-axis IMU sensor is mounted on the back of the hand portion of the glove. The MPU6050 provides:

- Three-axis accelerometer data (X, Y, Z) - measuring gravitational acceleration components for static orientation
- Three-axis gyroscope data (X, Y, Z) - measuring angular velocity for dynamic motion tracking

The MPU6050 is connected to the same I2C bus as the two ADS1115 modules, with the ESP32 managing all three devices over the shared bus.

The MPU6050's internal Digital Low-Pass Filter (DLPF) is configured directly via register-level programming using the Wire library on the ESP32. Rather than relying on default register settings, the DLPF register is explicitly written at startup to apply hardware-level noise filtering on both accelerometer and gyroscope outputs before the data leaves the sensor chip. Gyroscope sensitivity scaling is also configured via register programming to optimize measurement precision for the angular range of typical hand gestures.

Processing, Assembly and Output: An ESP32 dual-core microcontroller serves as the central processing unit. It manages:

- I2C communication with both ADS1115 modules and MPU6050
- Startup calibration routine
- All preprocessing and filtering
- Feature extraction and classification
- Serial communication with the laptop for output display and audio
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The complete circuit five op-amp boards, two ADS1115 modules, MPU6050, and ESP32 is assembled on a PCB/breadboard platform attached to the glove structure. A portable battery module powers the entire system, making it fully wireless during gesture capture.

The ESP32 communicates with a laptop via USB serial connection. The laptop receives the recognized gesture as text and plays the corresponding audio output currently supporting Indian language speech output. The system operates entirely offline with no internet connectivity required. A future version is planned as a mobile application where output text and audio are delivered directly to a smartphone over Bluetooth, removing the laptop dependency entirely.

Working of the Product

Power On and Startup IMU Calibration: When the glove is powered on, the system immediately enters a mandatory 5-second IMU startup calibration routine before any gesture recognition begins. During this period:

- The user holds the glove still in their natural comfortable resting hand position
- The system collects 2000 gyroscope samples across all three axes (X, Y, Z)
- The mean offset for each axis is computed:

$$\text{Offset} = \Sigma(\text{gyro readings}) / N, \text{ where } N = 2000$$
- These computed offsets are stored in memory and subtracted from all subsequent gyroscope readings throughout the session

A critical and novel aspect of this calibration is that it is fully position-independent. The system does not require the user to hold the glove at any specific fixed angle or predefined pose during calibration. Whatever natural resting angle the user holds their hand at when the glove is powered on becomes the calibration baseline. All subsequent orientation calculations and gesture predictions are made relative to this user-specific, session-specific reference frame.

This means:

- Different users with different natural hand resting angles are all accommodated automatically
- The same user calibrating from slightly different positions each session still gets consistent results
- No manual reset or repositioning is required between users

This position-independent startup calibration approach makes the system significantly more practical for real-world daily use compared to systems requiring a fixed calibration posture, and has not been specifically documented in this form in existing sign language glove prior art.

Gesture Capture: The user performs a gesture. All five flex sensors generate analog signals simultaneously. Each signal passes through its dedicated LM393 module individually potentiometer-tuned for that finger conditioning the signal before it reaches the ADS1115 modules. The ESP32 reads all five digitized values from both ADS1115 modules and simultaneously reads MPU6050 orientation data over the shared I2C bus.

Preprocessing: Gyroscope calibration offsets are subtracted from live readings. The complementary filter fuses gyroscope and accelerometer data into stable pitch and roll values:

$$\theta = 0.98 \times (\text{prev_angle} + \text{gyro} \times \text{dt}) + 0.02 \times \text{accel_angle}$$

This filter was chosen after testing Kalman, Madgwick, and Mahony filters — all added too much latency for real-time ESP32 operation. Each flex reading is then normalized into one of three discrete states per finger — straight, half-bent, fully bent — using individually calibrated per-sensor thresholds.

Feature Extraction and Classification: A 7-element feature vector is formed — five flex bend states plus pitch and roll angles. This is matched against stored OSL gesture templates in ESP32 memory using lightweight threshold-based pattern matching. When the vector falls within a template's tolerance range, that gesture is identified.

Output Generation: Once a gesture is successfully identified the ESP32 transmits the recognized gesture label to the connected laptop via USB serial communication. The laptop software:

- Displays the recognized gesture as readable text on screen
- Simultaneously generates and plays the corresponding speech audio output in the selected Indian language

The entire pipeline from gesture performance to text and audio output operates with low latency, fully offline, with no internet connection required.

Future Output Enhancement: A mobile application is planned where the ESP32 communicates recognized gestures directly to a smartphone over Bluetooth. The app will display output text and play audio on the phone, eliminating the laptop dependency entirely and making the system fully portable for daily use in educational institutions, healthcare environments, workplaces, and public spaces.

Own Sign Language (OSL)

The system includes a built-in Own Sign Language a custom designed sign language developed by our team specifically for this system. The complete OSL gesture set, including images of each letter sign, is documented in the accompanying research paper, allowing users to learn and use the language directly.

The OSL gesture templates are pre-stored in the ESP32's classification library. The system recognizes OSL letter and word gestures using the same feature vector and threshold-based classification pipeline described above, producing text and speech output in the user's selected Indian language.

7. Diagram

